



Predictive Model Plan

1. Model Logic (Generated with GenAI)

The predictive model is designed to estimate the probability that a customer will become credit delinquent based on historical financial behavior and demographic data. Using GenAI, I conceptualized a structured pipeline centered around *logistic regression*, a binary classification model ideal for predicting outcomes like delinquency (0 = non-delinquent, 1 = delinquent).

Model Workflow:

1) Data Ingestion

- Load customer records including features such as income, credit utilization, missed payments, and payment history.

2) Feature Engineering

- Encode categorical variables (e.g., employment status, location).
- Normalize numerical features and derive behavioural indicators.

3) Model Training

- Train logistic regression using selected features.
- Apply regularization (L1/L2) to prevent overfitting.
- Use cross-validation to ensure generalizability.

4) Prediction Output

- Generate probability scores for delinquency risk.
- Classify customers based on a decision threshold (e.g., 0.5).
- Output risk flags or scores for downstream decision-making.



Top 5 Input Features:

- Missed_Payments
- Credit_Utilization
- Debt_to_Income_Ratio
- Monthly Payment History (Month_1 to Month_6)
- Credit_Score

2. Justification for Model Choice

I selected **logistic regression** as the primary model due to its high interpretability, ease of deployment, and alignment with financial industry standards. In regulated environments like Geldium's, transparency is critical—logistic regression provides clear coefficients that explain how each feature influences risk. It's computationally efficient, scalable for large datasets, and produces intuitive probability scores that support decision-making. While more complex models like **Random Forests** offer higher accuracy in some cases, they lack the transparency required for compliance and customer trust. Logistic regression strikes the right balance between performance and explainability, making it ideal for Geldium's credit risk strategy.

Additionally, its simplicity allows for easier monitoring, auditing, and retraining, which are essential for maintaining model integrity over time.

3. Evaluation Strategy

To ensure the model is both accurate and fair, I propose a multi-layered evaluation strategy:



Performance Metrics:

- Accuracy: Measures overall correctness of predictions.
- Precision: Indicates how many predicted delinquents were actually delinquent.
- Recall: Shows how many actual delinquents were correctly identified
- F1 Score: Balances precision and recall, especially useful in imbalanced datasets,
- AUC-ROC: Evaluates the model's ability to distinguish between classes across thresholds.

Interpretation:

High accuracy alone isn't sufficient—if delinquency is rare, the model could be biased toward predicting non-delinquency. F1 score and AUC help assess performance more holistically. Precision ensures customers aren't unfairly flagged, while recall ensures risky customers aren't missed.

⚖️ Fairness & Bias Checks:

- Demographic Parity: Ensure equal positive prediction rates across groups (e.g., gender, region).
- Equal Opportunity: Check true positive rates across subgroups.
- Disparate Impact Ratio: Maintain fairness threshold (ideal range: 0.8–1.25).
- Stratified Performance Analysis: Evaluate metrics separately for protected groups.

⚡ Bias Mitigation (if needed):

- Reweighting samples to balance representation.
- Feature auditing to remove proxies for sensitive attributes.
- Post-processing adjustments to equalize outcomes across groups.



Ethical Considerations:

- Avoid over-penalizing vulnerable groups.
- Ensure model decisions are explainable and contestable.
- Monitor for drift and retrain periodically to reflect changing customer behaviour.
- Communicate risk scores responsibly to avoid undue harm or panic.